

RENANCE

FEDERAL UNIVERSITY, BIRNIN KEBBI

BUS 403 — Past Questions & Materials

by Renance

Course: BUS 403

Institution: Federal University, Birnin Kebbi

Papers in this pack: 1

Levels: 400

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Paper 1 of 1 — ANALYSIS FOR BUSINESS DECISION (Bus 403)

Level 400 • First Semester • Handout

2024/2025

BUA 403

ANALYSIS FOR BUSINESS DECISION

Department of Business Administration,
Faculty of Social and Management Sciences,
Federal University, Birnin Kebbi
2024/2025 Session

Learning Outcomes

At the end of this course, students will be able to:

- i. Explain the basic elements of decision analysis.
- ii. Demonstrate an understanding of operational research approach to business decision.
- iii. Apply optimization techniques to resource allocation.
- iv. Use different OR models to create and analyze the risk profile of a decision.
- v. Apply the knowledge of probability judgments to managerial decisions.
- vi. Apply Game Theoretic model to business decision situations
- vii. Apply Decision Tree model to business decision situations
- viii. Apply the concept of simulation to business decision situations.

COURSE OUT LINE

1. Introduction
2. The Concepts of Analysis and Decision
3. Domain of Analysis for Business Decision
4. Characteristics of Decision-Making Process
5. Structure of Managerial Decisions
6. Environment of Decision Making
7. Decision Tree Model
8. Game Theory
9. Simulation techniques
10. Sequencing, Routing and Line Balancing
11. Queueing Model

Learning Skills

- Business analytical skills: which enable students to approach business problems with logical thinking, and then to model and solve them using measurable data, quantitative methods, and critical reasoning.
- Critical thinking and problem solving skills: —the ability to make decisions, solve problems, and take action as appropriate

- Integrative learning skills: this teaches the students how to address real-world problems and opportunities across functional disciplines such as marketing, finance, accounting, management, and operations.

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Selected Texts

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Funso.A. (1996) Basic Operational Research Techniques Panaf Publishing Lagos

Howard .K. (1975) Quantitative Analysis For Planning Decision MacDonald & Evans Ltd London

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Monahan .G. E. (2000) Management Decision Making: Spreadsheet, Modelling, Analysis, and Applications. Cambridge university press United Kingdom.

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Rex et al (1974) Decision analysis: An overview. USA

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José Luis Bermúdez (2009) Decision Theory and Rationality, Oxford University Press, England

Charles Yoe (2019) Principles of Risk Analysis: Decision Making Under Uncertainty, 2nd Edition, CRC Press, New York

Rau´l Poler, Josefa Mula and Manuel Diaz-Madronero (2014) Operations Research Problems: Statements and Solutions, Springer, London

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1.0 INTRODUCTION

Decision making is certainly the most important task of a manager and it is often a very difficult one. Managers make decisions in complex situations. Complexity in the modern world, along with information quantity, uncertainty, and risk, make it necessary to provide a rational

decision-making framework. Analysis for Business Decisions (ABD) gives procedures for solving complex problems step by step. Its goal is to give guidance, information, insight, and structure to the decision-making process in order to make better and more 'rational' decisions. It is a process that allows the decision maker to select at least one option from a set of possible decision alternatives. It presents the decision-analysis process for business organisations, using different decision criteria, different types of methods and techniques that are sufficiently quantitative in nature. It describes the elements in the analysis of decision alternatives and choices, as well as the goals and objectives that guide decision-making.

It thus, provides the fundamental quantitative knowledge necessary for managers to improve on the quality of the decision-making process. This course will enhance students' ability to perform the quantitative analyses necessary, understand the usefulness and limitations of the methods; recognize situations where the methods can be applied beneficially and issues involved when utilizing the results of analyses.

2.0 CONCEPTS OF ANALYSIS AND DECISION

Analysis: This means the identification of components or the break down of a thing into different components and the study of each component in order to understand how each of them function individually. It also means, to identify and understand how they come together, interact and interrelate in order to achieve the objective of the whole. By interrelationship we mean the structure of the component. Part of this structure is the development process and how one leads to another. We try to understand what component exists? How do they behave? I.e. their relationship and how we can manipulate them.

The reason for analysis is to be able to make rational decision. Being rational is to be objective and logical, this enable us to achieve effectiveness and efficiency.

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2:1 CONCEPT OF DECISION

Decision: It may be defined as intentional and reflective choice in response to perceived needs. It means the final course of action one is committed to i.e. the choice one made out of several different alternatives. For example, if we have ABC as alternatives and we finally choose C, it means our decision is C. Thus, decision is judgment, which involves the choice of course of actions out of several other courses. Decision involves rigorous mental activities, and it is always directed towards solving problems. However, if there are no alternatives, there cannot be a choice and there cannot be decision.

When decision is made a decision-maker will be ready to hear all the consequences of such decision. A good decision is one that is based on logic, considers all available data and possible alternatives. A good decision may result in an unexpected or unfavourable outcome, but if properly made, it is still a good decision. A bad decision on the other hand, is not based on logic, does not use all available information, does not consider all alternatives, and does not employ appropriate quantitative techniques. If one makes a bad decision but he is lucky to get a favourable outcome it is still a bad decision. Thus, in order to achieve the best out of any decision situation, it is imperative to analyse all the alternative courses of action and expected

consequences of such alternatives.

Analysis for business decision can therefore be defined as the identification of the components of a decision situation using scientific approach and analysing such components so as to achieve the best and most rational decision. Decision Analysis is a discipline that addresses important practice of assessing and recommending course of action for the decision maker

2:2 CONCEPT OF RATIONALITY IN DECISION MAKING

Rationality means employing the best means to achieve the most worthwhile objective.

Rationality favours logic- (reasoning based on strict principles of validity), objectivity (lack of favouritism or freedom from bias), and analysis over subjectivity and insight. It assumes that people will make choices that maximize benefits and minimize any costs. That is people will choose the objective that provides the greatest reward at the lowest cost.

It involves two essential elements:

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a) The value system of the decision maker i.e. what is rational to Mr A may not be to Mr B.

b) Availability of information or infrastructure. This includes the level of information and the method of technology available at that point in time. Rationality is therefore a relative term and has to be considered on the bases of the two factors above.

Putting it slightly differently, personality traits affect a person's risk attitude and, therefore, influence how decisions are made. Personal biases may do the same. Ideally, the decision maker wishes to avoid bias but, inasmuch as our knowledge of the world is incomplete, pure, value-free "objectivity" is difficult to attain.

However, in order to reduce our level of biases, rationality does not consider factors that cannot be quantified. It leaves out consideration of personal feelings, loyalties, or sense of obligation. Its objectivity creates a bias toward the preference for facts, data and analysis over intuition or desires.

3:0 DOMAIN OF ANALYSIS FOR BUSINESS DECISION

One of the central demands of management of organizations is decision-making. In fact, it is often said that the essential role of management is decision making. As managers you must make decisions on a wide range of issues, in a wide range of context on a continuous basis. These decisions may range from simple to complex decisions or general to specific. Still some of these decisions involve long-term perspective and the restructuring of organizational process for the purpose of attaining the stated goals. Yet some are medium term whereas others are routine and recurrent in nature. Outcomes of decision-making are generally reflected in varieties of ways including those, which are regarded as policies, programmes, procedures, estimates, directives, plans, budgets etc.

Whatever the nature of decision and whatever the situation in which decisions are made,

managers must pay adequate attention to ensure not only that these decisions are accurate and reliable but also appropriate and effective. Analysis for business decision describes a number of techniques, mostly quantitative in nature, that provide efficient tools for the improvement in the accuracy and quality of business decisions. Since it's generally assumed that the quality of

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an organization is known better by the quality of its decisions, analysis for business decision has continued to assume better prominence in the set of tools utilized and skills acquired by managers.

Analysis for Business Decision otherwise refers to as Management Science, Quantitative Techniques, Quantitative Analysis or Operation Research assist managers by providing an approach and a framework for managerial decision making. The basic framework procedure involves the following: (1) Observation (2) problem definition (3) model specification (4) model solution (5) implementation.

(1) OBSERVATION

The first step in management science is to observe the environment whether or not there is problem, which needs solution. This is essentially in two respects:

- (a) To re-establish the fact that problem that calls for decision-making intervention is rooted in the environment; in other words, there is an environmental context for every decision.
- (b) It shows that problems addressed are geared towards improving condition or situation that currently exists. The environment of decision includes: The internal and external environment. The internal environment includes the six Ms namely; manpower, money, machinery, material, market and method. The external environment includes the technology, society or culture, policies, economic situation and legal system among others. Having made observations, the specific nature of the problem has to be identified and this is regarded as problem definition.

(2) PROBLEM DEFINITION

Organisational problems are often as difficult to identify as medical ailments. Common symptoms of management problems are declining profits, escalating costs, employee absenteeism, poor quality products, conflicts, and employee stress. Frequently, however, there are no apparent symptoms, and managers must be on the alert to discover potential problems. Personnel managers for example, track performance through formal periodic evaluations to detect early signs of poor employee performance. Quality control managers regularly monitor product performance. Market researchers gather information from customers to better understand the competition. Other diagnostic efforts range from political observations to machine maintenance inspections.

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Symptoms provide clues to existing or potential problems, and when they are diagnosed correctly, managers can accurately define the right problems to solve. The diagnostic process often leads to much more than problem solving. If an organisation has a good network of sensors (feedback reports, quality control checks, and market research data), it will also be able

to uncover new opportunities, such as unmet consumer needs.

To define the problem means much more than simply providing a written description of it. It is to establish its boundaries. This involves locating and isolating it as well as articulating it. This helps to specify the scope and limit of the decision-making problem so as to ensure that focus is kept strictly on the problem and not anything else.

According to Kepner and Tregoe (1981) problem definition requires cause-and-effect thinking. A problem is the visible effect of a cause that resides somewhere in the past. We must relate the effect we observe to its exact cause. Only then can we be sure of taking appropriate corrective action-action that can correct the problem and keep it from recurring. Steps include describing the problem, identifying and evaluating possible causes, and testing possible causes to find true cause before corrective actions are taken.

Description of the Problem can be done in four dimensions as follows:

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- i. IDENTITY - what it is we are trying to explain
- ii. LOCATION - where we observe it
- iii. TIMING - when it occurs
- iv. MAGNITUDE - how serious, how extensive it is

All obtainable information about any problem will fall within one of these four dimensions. Within each we ask specifying questions that will flesh out our description of the effect of the problem and give us exactly the kinds of information that will be most useful for analysis.

Often, decisions are swayed by sentiments or emotions and by other considerations that tend to prevent the decision making process. Therefore, it helps to establish the current dimension and ramification of decision-making problem.

(3) MODEL SPECIFICATION

Once the decision-making problem has been carefully defined the stage is reached for modelling which helps to create a make- believe scenario or an abstract formulation of functional relationship that is relevant not only to the analysis of the problem, but also appropriate manipulation to obtain a solution.

Simply described, a model is representation of reality. The use of model is perhaps the most important distinguishing feature of the techniques used in analysis for business decision.

Although a model may be descriptive, theoretical, graphical or quantitative, the last (i.e. quantitative models) are invariably the ones employed in analysis for business decision.

A model does not necessarily have to be complex or highly sophisticated to be relevant in analysis for business decision. What is required is that it must adequately and effectively represent reality in a way that makes manipulation possible and solution relevant and functional. In this regard simple equations and formulae qualify as models in their own respects. Some of the models used in analysis for business decision include; linear programming, decision trees, mathematical programming, line balancing, sequencing, queuing

etc.

(4) MODEL SOLUTION

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This refers to the manipulation and operation performed to use a given model to solve a given problem. Once the model has been carefully specified both by variables and by parameters, it is generally straightforward to manipulate it to obtain solution.

(5) IMPLEMENTATION

This involves the application of the specific model solution to solve the problem at hand.

3:1 USES OF QUANTITATIVE ANALYSIS IN BUSINESS MANAGEMENT

ABD is designed to assist management and businessmen in the following ways:

1. It helps in breaking down complex problems into smaller and easier parts to handle.

The key word is analysis, which refers to the process of breaking something down into its constituent parts. Decision analysis therefore involves the decomposition of a decision problem into a set of smaller (and, hopefully, easier to handle) problems. After each smaller problem has been dealt with separately, decision analysis provides a formal mechanism for integrating the results so that a course of action can be provisionally selected. This has been referred to as the 'divide and conquer orientation' of decision analysis. So, decision analysis helps to appreciate the complex nature of business decision and the need for thorough analysis of each problem situation.

2. It helps decision maker make clear and explicit judgments

Because decision analysis requires the decision maker to be clear and explicit about his or her judgments, it is possible to trace back through the analysis to discover why a particular course of action was preferred. This ability of decision analysis to provide an 'audit trail' means that it is possible to use the analysis to produce a defensible rationale for choosing a particular option. Clearly, this can be important when decisions have to be justified to senior staff, colleagues, outside agencies, the general public or even oneself.

3. It can lead to greater understanding and communication of the choice made

When there are disagreements between a group of decision makers, decision analysis can lead to a greater understanding of each person's position so that there is a raised consciousness about the issues involved and about the root of any conflict. This enhanced communication and

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understanding can be particularly valuable when a group of specialists from different fields have to meet to make a decision. Sometimes the analysis can reveal that a disputed issue is not worth debating because a given course of action should still be chosen, whatever stance is taken in relation to that particular issue.

4. It can lead to stakeholders' commitment

Decision analysis allows the different stakeholders to participate in the decision process and

develop a shared perception of the problem, it is more likely that there will be a commitment to the course of action which is eventually chosen.

5. It can lead to creative thinking

Decision-analysis approach gives great insights into decision situations, as such creative thinking may result so that new, and possibly superior, courses of action can be generated. The analysis can also provide guidance on what new information should be gathered before a decision is made. So, the purpose of decision analysis is to produce insight and promote creativity to help decision makers

make better decisions.

6. It is conditionally prescriptive

While we should not expect decision analysis to produce an optimal solution to a problem, the results of an analysis can be regarded as being 'conditionally prescriptive.' By this we mean that the analysis will show the decision maker what he or she should do, given the judgments which have been elicited from him or her during the course of the analysis. The basic assumption is that of rationality. If the decision maker is prepared to accept a set of rules (or axioms) which most people would regard as sensible then, to be rational, he or she should prefer the indicated course of action to its alternatives.

4:0 CHARACTERISTICS OF DECISION-MAKING PROCESS

Decision making process has the following essential characteristics:

(1) Decision - Maker: This is the person who will finally be committed and responsible for the consequences of the decision made. It is possible for the decision -maker to consult analysts or consultants during the decision-making process but they are not the decision-makers. They are only to recommend for the decision-maker. The decision-maker can be an individual or cooperate or business entity.

(ii) Objective or the problem: - we have to identify the objective that is to be achieved by the decision. This is because our decision has to be made in relation to the objective to be
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achieved. Objective should be distinguished from mere symptoms. Indication should be given on how to measure the objective and measure its extent of achievement. Our job in this case is to help to quantify the objective, which makes measurement easier.

(iii) Generating alternatives or options: these options are the available alternative ways of achieving the same objective. Options are generated by gathering information exhaustively. The possible options have to be exhaustive or else we shall be making decision that will be invalidated since there may be better options not yet exploited, therefore there should be creativity in options generation. Approaches such as lateral thinking, consulting experts, brainstorming etc should be adopted when generating options.

(iv) Evaluation Criteria: After generating options we have to evaluate (score) them so as to obtain their relative value. This is done by using evaluation criteria. Such evaluation criteria have to be identified e.g. if one objective is to employ a competent medical doctor, the options are the various applications. To know their relative worth, we may use evaluation

criteria such as qualification, experience, age, social status etc.

(v) Evaluation of options: After identifying the criteria to be used, we then use them appropriately; this involves scoring each of the various options over each of the various criteria to obtain their relative worth. Most times we use screening and scoring methods.

Using appropriate methods, alternatives are evaluated on three criteria: sufficiency, feasibility, and realism.

(vi) Choice- making: After evaluating and obtaining relative value of the options we then make our decision. The choice is made in terms of what the objective specifies. An objective may say choose the candidate with the highest score. Another objective may say choose any candidate scoring above certain range. Ideally, a manager should make a decision that is optimal. However, constraints often lead to satisficing (choose less-than-optimal). If a problem has been accurately formulated and sufficient viable alternatives exist, then a manager should, make a choice that is best under the circumstances. This will be the optimal choice—one that generates the greatest possible benefits with the fewest negative consequences.

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